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ORIGINAL RESEARCH PAPER

A generalised comparison of Pareto/NBD based forecasts using MCMC, maximum likelihood, and heuristics

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Abstract
This study is the first generalised effort to compare the forecasting efficacy of different Pareto/NBD model-based forecasts. Monte Carlo Markov Chain (MCMC)-based estimates, Maximum Likelihood Estimation (MLE), and heuristics are applied to four different types of forecasts: (1) predicting the future number of purchases a single customer makes within a given period, (2) identifying active customers who will make at least one purchase within a given period, (3) identifying the customers who will belong to the top segments of the customer base, and (4) predicting the timing of a customer's next purchase. The results show that the model-based forecasts outperform the heuristics regarding predictive power and accuracy for the first three types of forecasts. MCMC yields slightly better results than MLE and it can additionally convince with confidence intervals for the number of future purchases. Forecasting the timing of a customer's next purchase yields deviations that are too large to be used in practice.

Keywords Customer base analysis · Pareto/NBD model · Markov Chain Monte Carlo

JEL classification C11 · C15 · C52 · C53 · C63

1 Introduction

The Pareto/Negative Binomial Distribution (Pareto/NBD) model (Schmittlein et al. 1987) was the first buy-till-you-die (BTYD) model in the non-contractual setting of customer base analysis (CBA). These models assume that a customer continues buying until an unobserved dropout occurs at which the customer becomes permanently inactive. The purchase process of a BTYD model describes the purchase pattern of a customer, whereas the dropout process defines the distribution of the dropout time.

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Its main deficiency at the time was its high computational effort, leading to the development of simplified models like the (M)BG/NBD (Fader et al. 2005a; Batislam et al. 2007). These models share the NBD purchase process but use a (modified) beta geometric dropout process instead, allowing the Maximum Likelihood estimates (MLE) to be determined using spreadsheets (Fader et al. 2005b). Today's system capacities enable a low threshold use of model extensions such as the consideration of both static and time-variant covariates (Bachmann et al. 2021). Subsequent models make the assumptions more flexible by assuming gamma distributed rather than exponentially distributed inter-purchase times (Platzer and Reutterer 2016) or simultaneously analysing different cohorts (Gopalakrishnan et al. 2017). These sophisticated models require Markov Chain Monte Carlo (MCMC) algorithms as closed-form likelihoods are unavailable, but researchers also provide ready-to-use packages for their application (Platzer 2021a; Bachmann et al. 2022) to account for the higher computational complexity.

In recent years, the trend towards machine learning (ML) has also spread to customer base analysis. The fundamentally different approaches of parametric models and ML routines each have their strengths and weaknesses. In general, well-fitting parametric models can play to their strength of coping well with parsimonious data sets due to their pre-defined structure, whereas ML algorithms can recognise patterns and trends that are not captured by parametric models.

Valendin et al. (2022) showed that their RNN-based model has a higher predictive accuracy at the individual level and a lower bias at the cohort level than the established parametric models. Xie (2020) compared different ML algorithms with the predictions of the Pareto/NBD model. The results show that the parametric model is superior in predicting transaction frequency, while it performs weaker in predicting inactivity. It also benefits from a short calibration length and a long forecast period.

Besides these data set specific results, there are some general aspects to consider. As both Valendin et al. (2022) and Xie (2020) pointed out, ML routines cause significantly more computational cost than parametric models. Another structural advantage of parametric models over ML is the type of output which contains interpretable parameters, predictions, and, in case MCMC is used, confidence intervals for both.

Alternatively, parametric models can also be estimated via Maximum Likelihood Estimation (MLE), which is computationally more efficient than MCMC, but initially only yields point estimates. To account for uncertainty in this case, confidence intervals can be approximated using bootstrapping. Notably, applying bootstrapping to MLE models involves substantially lower computational and memory requirements than in ML. This is primarily because bootstrapped ML approaches require retraining full models for each resample, whereas in parametric settings, only parameter re-estimation is necessary.

The fundamental choice between a parametric or machine learning (ML) model in CBA depends on various factors, requiring careful consideration of aspects such as the specific application, the available data, and computational resources. Ultimately, parametric models will continue to have their place in CBA.

Even though there are closed form solutions for the Pareto/NBD model, different MCMC implementations have been developed in the past (Ma and Liu 2007; Abe

2009; Singh et al. 2009). The investigation of the different estimation procedures may seem like a purely academic question, but it is worth taking a closer look at them.

Thinking of the likelihood as an unnormalised probability distribution, Maximum Likelihood calculates its mode, i.e., the parameter value(s) yielding the highest likelihood value. However, MCMC determines the complete posterior distribution and hence enables the derivation of its mean, median, and quantiles. In contrast to MLE, MCMC not only provides the full posterior distribution for the heterogeneity parameters, but also for the individual customers' parameters. This much wider range of available information suggests that more and better predictions could be made using MCMC instead of MLE. Simon and Adler (2022) have shown that the parameter recovery as well as the individual customers' purchase forecast improves significantly when using the median draw of Abe's (2009) MCMC algorithm rather than MLE, particularly in the context of small sample data sets.

Newly developed CBA models usually provide formulas for the expected number of future purchases and the probability of a single customer still being active at a specific time T (called P(alive)). The accuracy of the purchase forecast is the well-established benchmark measure to compare the goodness-of-fit of different models. As the duration of the customer relationship with a company is not observable in a non-contractual setting, P(alive) is difficult to assess for real data sets and hence infeasible for model comparison.

The aim of CBA is to decide which customers are worth being individually targeted and how this targeting can be customised, based on the customer's previous purchase history. The first of these objectives is mainly driven by the concept of customer lifetime value (CLV), which is given by the expected (discounted) future cash flows from the customer relationship (Gupta et al. 2006) and can be based on the parameter estimates from the Pareto/NBD model (Glady et al. 2009; Abe 2009; Bachmann et al. 2021). Tailoring marketing activities like individual mailings, coupons, or special offers relies on purchase forecasts which, in case of MLE estimates, are based on cohort rather than individual parameters.

To optimise the budget allocation, the following types of predictions are of particular importance:

(1) predicting the individual number of purchases made in a given forecast period,

(2) identifying customers which make at least one purchase within a given forecast period ("active" customers),

(3) identifying the Top 10% and Top 20% of the customer base,

(4) predicting the timing of the next purchase.

Practitioners often rely on heuristics as they are easy to implement. They may feel confirmed by previous studies like Wübben and Wangenheim (2008) who compared the performance of BTYD models and heuristics regarding the prediction of the purchase level, the activity status, and the future best customers. Their study is based on three empirical data sets and claims that the heuristics perform at least as well as the stochastic models. At a similar time, but without the consideration of heuristics, Batislam et al. (2007) assessed the purchase and activity prediction of the Pareto/NBD model and the (M)BG/NBD model using one empirical data set and obtained similar results for all the models. Huang (2012) performed a simulation study to compare the performance of the customer prioritisation for the Pareto/NBD model and a heuristic. In his study, the heuristic outperforms the model-based forecast under certain conditions. Korkmaz et al. (2013) compared predicting the timing of the next purchase for different BTYD models and relied on three empirical data sets.

Despite these various contributions, there are still major research gaps left for each of the aforementioned types of predictions which are being addressed in this paper. A summary is given in Table 1.

In the literature to date, the *future number of purchases* has been used more as a means to an end in order to compare the goodness-of-fit of different models based on a handful of empirical data sets (Fader et al. 2005a; Bachmann et al. 2021; Bemmaor and Glady 2012; Platzer 2021b; Valendin et al. 2022). Simon and Adler (2022) reported point estimates for the deviations measures in their simulation study, but their contribution neither contains empirical validation nor confidence intervals. Still, the magnitude of the expected deviation is of major interest for practitioners but has not yet been subject of research.

Previous studies (Simon and Adler 2022; Batislam et al. 2007; Wübben and Wangenheim 2008; Schmittlein et al. 1987) define *active customers* solely based on their dropout process, regardless whether or when they are going to make another purchase in the future. However, even though this may be correct from a theoretical standpoint, this definition is misleading in practice. If a customer drops out in a few months without making another purchase, he or she is still classified as active. The same applies to a customer who makes his or her next purchase in the distant future, which is outside the company's planning horizon. Neither of these two consumers corresponds to the common understanding of an active customer and should therefore not be treated as such. This is supported by Reinartz and Kumar (2000), who have shown that customers with a long lifetime are not necessarily the profitable ones. As the lifetime is not observable, another problem arises as companies cannot validate how well this forecast works for their customer base. Hence, this paper re-defines as customer as active if he or she makes a purchase within a given time period (e.g., the company's planning horizon) which is practically relevant and also allows validation. The analysis of this new type of classification also compares different ways of predicting the activity status based on MLE- and MCMC parameter estimates.

Regarding customer prioritisation (i.e., identifying the *Top A%* of the customer base), Huang (2012) performed a simulation study comparing Pareto/NBD distributed data sets with a heuristic. He omitted the parameter estimation procedure and used the data set generating parameters instead. Even though this facilitates the computational effort of such a simulation study, it reduces the transferability to real-world data sets. In a second contribution, Wübben and Wangenheim (2008) also analysed the customer prioritization of three empirical data sets comparing MLE-based parameter estimates for the Pareto/NBD model with a heuristic. However, their methodology contains a bias in favour of the heuristic. CBA data sets of consumer goods tend to have a majority of customers with just a few purchases, leading to many customers sharing the same number of observed purchases. As the authors' heuristic assumes that the previous Top A% are going to be the future Top A%, these multiple values allow far more than A% of the

Table 1 Summary of previous studies and resulting research gaps

Prediction type	References	Simulated/empirical	Shortcoming/research gap
Future purchases	Batislam et al. (2007)	Empirical (1)	Not generalisable due to the small sample size, no consideration of MCMC
	Wübben & Wangenheim (2008)	Empirical (3)	
	Simon & Adler (2022)	Simulated	No confidence intervals, no empirical validation
Active customers	Batislam et al. (2007)	Empirical (1)	Active customers are defined by the timing of their dropout only, neglecting their future purchase behaviour, no consideration of MCMC
	Wübben & Wangenheim (2008)	Empirical (3)	
Top customers	Huang (2012)	Simulated	Use of data set generating parameter rather than estimates, no empirical validation
	Wübben & Wangenheim (2008)	Empirical (3)	Methodological bias in favour of the heuristic due to multiple values, not generalisable due to the small sample size, no consideration of MCMC
Next purchase	Korkmaz et al. (2013)	Empirical (3)	Definition of the deviation measure causes a severe methodological bias, not generalisable due to the small sample size, no consideration of MCMC

Note: The numbers reported for the empirical studies state the number of real-world data sets analysed customers to be assigned to the top A% segment. Since this set of predicted top customers is significantly larger than the set from the more granular parameter-based classification, the apparently high accuracy of the heuristic for identifying the Top A% of the customer base is misleading. To increase the granularity of the heuristic in this paper, we do not use the number of purchases, but the purchase frequency which additionally considers the lifetime with the firm. In accordance with the other types of prediction, we also compare different parametric approaches based on MLE and MCMC which has not been done before.

Korkmaz et al. (2013) were the first to assess the prediction of the *timing of the next purchase*. Besides them using MLE estimates only, their paper suffers from a major methodological weakness. The customers of interest in this forecast are those who make an (observed) purchase in the future. The deviation measures should therefore only refer to them. Instead, for customers who make no (observed or predicted) purchase within the prediction period, the authors assign the end of the observation period as their (hypothetical) purchase date. This leads to a vast underestimation of the deviation measures, particularly in data sets with many inactive customers, as all of these contribute a deviation of zero. In this paper, the deviation is measured for those customers only who make an observed purchase within the forecast period. In addition, the timing of the next purchase is being predicted using MLE, MCMC, and a heuristic which is an extension to Korkmaz et al. (2013) who only considered MLE.

This simulation study with 3,000 data sets fills the research gaps that were discussed above by systematically investigating the magnitude for the different predictions. In order to obtain a comprehensive picture, the analysis uses a variety of parameter estimates. In addition to the MCMC estimators, these also include the MLE point estimates as well as the median and mean values based on MLE, which can be determined by bootstrapping. In a second step, the results of the simulation study are validated using four real data sets by examining whether the results of the deviation and accuracy measures lie within the intervals determined in the simulation study.

The remainder of this paper is organised as follows. The first section is a recap of the Pareto/NBD model assumptions and comprises the management questions and the derivation of the corresponding predictions and their goodness-of-fit measures. If applicable, these performance measures are determined based on different estimation methods (MCMC, MLE, and heuristics). The second section introduces the data base of the analysis which consists of the simulation study framework and the four empirical data sets. The third section presents the results of the analysis, the limitations and implications are discussed in the last section.

2 Forecasting methods

The Pareto/NBD model is based on the NBD model (Ehrenberg 1959) which assumes that the inter-purchase times (ipt) of the i-th customer are exponentially distributed with parameter λ_i, where the parameters $\{\lambda_i\}_{i=1,\ldots,n}$ follow a Gamma distribution with parameters r and α, i.e.

$$\{\lambda_i\}_{i=1,\ldots,n} \sim \gamma(r, \alpha). \tag{1}$$

Schmittlein et al. (1987) enhanced this model by adding a latent attrition process to account for the non-observable duration of the customer relationship with the firm. This customer "lifetime" $\{\tau_i\}_{i=1,\ldots,n}$ is assumed to be exponentially distributed with parameter μ_i, where $\{\mu_i\}_{i=1,\ldots,n}$ follow a Gamma distribution with parameters s and β, i.e.

$$\{\mu_i\}_{i=1,\ldots,n} \sim \gamma(s, \beta). \tag{2}$$

The classical Pareto/NBD model introduced by Schmittlein et al. (1987) assumes independence between the purchase and dropout processes, a property that simplifies the parameter estimation and has made it a widely accepted benchmark in the field. Relaxing this assumption would enable frequent buyers to have a longer lifetime with the firm. Abe (2009) allowed a dependency between the purchase and the dropout parameters, but his empirical analysis confirmed the independence assumption across multiple datasets. Given that the classical model has been validated in practice and allows for estimation via both MLE and MCMC, it remains the appropriate choice for this study's comparative analysis of estimation methods.

The heterogeneous likelihood L(r, α, s, β | data) of the Pareto/NBD model can be written in closed form and therefore allows traditional parameter estimation using MLE, which only provides point estimates for the heterogeneity parameters {r, α, s, β}. Confidence intervals for the MLE can still be be determined using bootstrapping.

Simon and Adler (2022) have shown that the MCMC implementation introduced by Abe (2009) outperforms the other procedures (Singh et al. 2009; Ma and Liu 2007) regarding the parameter recovery and the purchase forecast accuracy which will therefore be used in this study. MCMC provides full posterior distributions not only for the heterogeneity parameters {r, α, s, β}, but also for the individual parameters $\{\lambda_i, \mu_i\}_{i=1,\ldots,n}$. As Abe (2009) uses a data augmentation technique which simulates each customer's dropout time $\{\tau_i\}_{i=1,\ldots,n}$ in this non-contractual setting, these values can additionally be used to help making individual predictions. The details of the MCMC procedure are described in Sect. 7.

2.1 Purchase forecast

Predicting the number of future purchases $\{x^*_i\}_{i=1,\ldots,n}$ for a given forecast period (T, T + T*] (where T denotes the end of the estimation period) is a major aim of CBA and classically serves as a benchmark to compare different models. Still, practitioners lack a guidance on how accurate these predictions are. Hoppe and Wagner (2010) and Simon and Adler (2022) deduced minimal data set requirements for the application of the Pareto/NBD model, but these refer to the parameter recovery rather than the forecast accuracy. This study fills this research gap by contributing a magnitude for the deviation of the purchase forecast. In doing so, two different MCMC-based purchase forecast methods, the closed form MLE forecast and a management heuristic (Wübben and Wangenheim 2008), are compared with each other. All four methods are described in detail in 7.2.

The first purchase forecast $\{x^*_{fc,i}\}_{i=1,\ldots,n}$ based on MCMC uses *simulated purchase patterns (SPP)*. For customers who are supposedly alive (i.e., $\tau_i > T$), future purchases are simulated by drawing ipts from an exponential distribution $Exp(\lambda_i)$ until the end of the forecast period has been reached or the customer has dropped out, whichever occurs first. Customers who have supposedly dropped out already (i.e., $\tau_i \le T$) are being assigned a purchase forecast of zero.

The second MCMC-based purchase forecast $\{x^*_{fc,i}\}_{i=1,\ldots,n}$ uses the *individual conditional expectation (ICE)*. Fader et al. (2005a) provide a closed form expectation (see (*11*)) for the number of future purchases given that the customer has not dropped out yet. Again, the lifetime estimates $\{\tau_i\}_{i=1,\ldots,n}$ can be used here. As in the SPP procedure, customers who are deemed to have dropped out (i.e., $\tau_i \le T$) receive a purchase forecast of zero.

Since MLE provides estimates for the heterogeneity parameters $\{r, \alpha, s, \beta\}$ only, the individual approaches used for MCMC are not applicable here. Schmittlein et al. (1987) provide a *heterogeneous formula* for the expected value of the future purchases where the individual parameters are being integrated out by taking the expectation over the joint posterior distribution of $\{\lambda_i\}_{i=1,\ldots,n}$ and $\{\mu_i\}_{i=1,\ldots,n}$. The result is the closed form expectation (*12*) which uses $\{r, \alpha, s, \beta\}$ instead.

The *heuristic* simply assumes that every customer keeps buying at his or her previous mean purchase rate (Wübben and Wangenheim 2008).

To fully comprehend the different types of purchase forecasts, it is important to be aware of some structural differences. The MCMC-based prediction methods are being applied to each MCMC draw, giving a full posterior distribution for the purchase forecast. It is common to use its median as a point estimate to reduce the impact of outliers. This also holds for this study unless something else is stated. As SPP and ICE both use the dropout dates $\{\tau_i\}_{i=1,\ldots,n}$, all customers who are classified as dropped out in more than 50% of the MCMC draws will obtain a forecast of exactly zero as their median purchase prediction is zero. In contrast to this, the MLE-based expectation always includes a non-zero probability that the customer is still alive and is therefore always strictly greater than zero. The heuristic also yields strictly positive predictions since it does not contain a dropout concept. In addition to this, the SPP method has the unique feature of providing integer purchase forecasts. This is because the simulation process generates an integer prediction in each MCMC draw which also leads to an integer median value.

We use different fit measures to assess the individual purchase forecast. The mean absolute error (MAE), the median absolute error (MdAE), and the root mean squared error (RMSE) are being normalised wrt to the different purchase intensities across the data sets by dividing them by the average number of purchases made within the forecast period, leading to the definitions below. Since many customers make no purchase at all within the forecast period, relative measures like the median absolute percentage error (MAPE) cannot be computed as this would require division by zero.

If $\{x^*_{true,i}\}_{i=1,\ldots,n}$ denotes the true number of repeat purchases in the forecast period, the normalised mean absolute error (nMAE) is defined as

$$\mathrm{nMAE} = \frac{\sum_{i=1}^{n} \left| x^*_{fc,i} - x^*_{true,i} \right|}{\sum_{i=1}^{n} x^*_{true,i}}. \tag{3}$$

The normalised root mean squared error (nRMSE) penalises large deviations disproportionally which makes it a valuable supplement to the nMAE. It is defined as.

$$\mathrm{nRMSE} = \frac{\sqrt{\sum_{i=1}^{n} \left(x^*_{fc,i} - x^*_{true,i} \right)^2}}{\frac{1}{n}\sum_{i=1}^{n} x^*_{true,i}}. \tag{4}$$

The normalised median absolute error (nMdAE) is closely related to the nMAE but uses the median deviation rather than its mean:

$$\mathrm{nMdAE} = 100 \cdot \left| \frac{x^*_{fc,i} - x^*_{true,i}}{\frac{1}{n}\sum_{i=1}^{n} x^*_{true,i}} \right|_{0.5}. \tag{5}$$

2.2 Active customer identification

Besides the purchase forecast, identifying active customers is the second forecast feature of BTYD models (Schmittlein et al. 1987; Batislam et al. 2007; Wübben and Wangenheim 2008). They define a customer active if he or she has not dropped out until the end of the observation period, i.e., if $\tau_i > T_i$. As pointed out before, this understanding of an active customer does not meet the requirements for allocating scarce marketing resources if the focus lies on the short and mid-term customer relationships.

To overcome these deficiencies, we define a customer active if he or she makes at least one purchase in $(T, T+T^*]$. Unlike the purchase forecast, this is not about *how often* a customer will buy, but *if* he or she will buy at all within the prediction period. Customers are classified active if their purchase forecast $\{x^*_{fc,i}\}_{i=1,\ldots,n}$ from the four methods (SPP, ICE, MLE, and heuristic) is at least $c = 0.5$. In addition, the impact of choosing different threshold values c needs to be examined.

The sensitivity, precision, and overall accuracy are used as fit measures for the classification:

$$\mathrm{Sensitivity} = \frac{\#\text{correctly classified repeat buyers}}{\#\text{true repeat buyers}}, \tag{6}$$

$$\text{Precision} = \frac{\text{\#correctly classified repeat buyers}}{\text{\#predicted repeat buyers}}. \tag{7}$$

$$\text{Overall accuracy} = \frac{\text{\#correctly classified customers}}{\text{\#customers}} \tag{8}$$

2.3 Future Top A% customers

The top customer segment of a company plays a crucial role for a business' success and revenue generation. These customers are typically loyal, spend more on products or services, and contribute significantly to the company's bottom line (Mulhern 1999). Therefore, prioritising the correct customers is essential for enabling disproportionate marketing strategies.

Unlike the repeat buyer classification where the size of the target group depends on the customer base characteristics and the length of the calibration period, this is about identifying a given ratio (10% and 20%) of the most profitable customers. For ordering the customers, the CLV (Malthouse and Blattberg 2005; Jasek et al. 2018) or the purchase forecast $\{x^*_{fc,i}\}_{i=1,\ldots,n}$ (Wübben and Wangenheim 2008; Huang 2012) can be used as reference values. To make the determination manageable without requiring additional information (like purchase amounts, revenues, or discount factors), this study relies on $\{x^*_{fc,i}\}_{i=1,\ldots,n}$.

Huang (2012) performed a simulation study on the accuracy of the customer prioritisation for the Pareto/NBD model using the data set generating parameters, whereas Wübben and Wangenheim (2008) used $\{x^*_{fc,i}\}_{i=1,\ldots,n}$ based on MLE. For each of the methods described in 2.1, the Top A% ($A \in \{10; 20\}$) of the customer base are identified by ordering them by their $\{x^*_{fc,i}\}_{i=1,\ldots,n}$. Even though the classification and its evaluation seem to be very straightforward, they require taking a closer look at the details.

Typical datasets in CBA contain many consumers who only make a few purchases. This leads to several customers sharing the same number of actual future transactions $\{x^*_{true,i}\}_{i=1,\ldots,n}$ and impedes a clear cut-off at the bottom of the A% because of these multiple values. Hence, the Top A% segment may (and in many cases also will) contain more than A% of the customer base. To a certain extent, this inaccuracy is not problematic. However, if we e.g. have a case where the 80% quantile does not identify 20% of customers, but 50% due to a lack of diversification in the future purchases, then the goal of customer prioritisation cannot be achieved. In most cases, the cause lies in very low overall purchase rates in the data set, so that customer prioritisation is not only not possible here, but also does not make sense.

For this reason, only those data sets from the simulation study are used where at least A% of the customers actually make a purchase in the forecast period to ensure that the (100 – A)% quantile is greater than zero. This reduces the data base from 3,000 to 2,309 data sets for the Top 10% and to 1722 data sets for the Top 20%. In case of multiple occurrence of the quantile threshold, all customers with the same

$x^*_{true,i}$ as the (100 – A)% quantile are included, which also extends the corresponding prediction list to the same length. For example, if 24% of the customers in a data set belong to the Top 20% segment due to multiple values, we compare these with the predicted Top 24% to compare equally large sets and avoid bias between the methods.

The problem with multiple values also arises in the SPP prediction of the future purchases because this is the only method which obtains integer forecast values. To avoid this problem, the mean rather than the median value of the individual purchase prediction is used here as it gives us quasi-continuous values, allowing a unique order for the customer priority.

Depending on the definition of the heuristic, there is also a risk of multiple values occurring here. The variant proposed by Wübben and Wangenheim (2008) assumes the previous Top A% to be the future Top A%. This refers only to the number of observed purchases, which makes it difficult to differentiate between customers, especially in data sets with a low purchase rate. This leads to the aforementioned bias in favour of the heuristic, as the number of customers belonging to the top segment according to the heuristic is much larger than both the set from the parametric segmentation and the set of the actual top customers.

To mitigate this effect, we use the purchase frequency as a heuristic instead of the absolute number of purchases, which also includes the individual observation period. This means that a customer has a higher purchase frequency if he or she made the same number of purchases in a shorter time period because the initial purchase occurred later than that of another customer.

2.4 Timing of the next purchase

Predicting the timing of the next purchase $t_{x+1,i}$ can be a powerful tool as it may enable an on time customer contact. Still, Korkmaz et al. (2013) were the first to raise and apply this idea to three empirical data sets using MLE estimates and a Metropolis–Hastings algorithm to draw the individual parameters from the posterior distribution. The main difficulty in predicting the timing of the next purchase is that churned customers do not make a repeat purchase and therefore don't provide a value for $t_{x+1,i}$. In addition, repeat buyers with a very long (true or estimated) inter-purchase time extremely bias the measures of deviation. Korkmaz et al. (2013) approached this problem by defining a forecast period (T, T + T*] and assigning the value (T + T*) to all those customers who made no (real or predicted) purchase within this interval. Unfortunately, aggregated measures like the MAE suggest a very small deviation when the proportion of non-buyers in the prediction period is large.

To overcome this deficiency and to ensure that every forecast has a proper reference value, we only consider those customers with an actual purchase in (T, T + T*]. Omitting inactive customers can significantly reduce the customer base, but at the same time it means that the error measure is limited exclusively to the group of (active) customers that are of interest to us. Since this approach assumes that the customers under consideration are active until the next purchase, we can disregard the dropout process for the forecast of t_{x+1}.

Because of the memoryless property, the distribution of the timing of the next purchase can be described as $t_{x+1,i} \sim T_i + Exp(\lambda_i)$. The MCMC based prediction of $t_{x+1,i}$ can easily be derived by imputing the median estimate of λ_i since the exponential distribution is strictly monotone. The MLE-based value for $E(t_{x+1} | x,T,r,\alpha)$ does not converge for single buyers if $r < 1$. Hence, the median $Q_{50}(t_{x+1} | x,T,r,\alpha)$ is used to ensure finite estimates. The derivations can be found in 7.3.

The heuristic for $\{t_{x+1,i}\}_{i=1,\ldots,n}$ assumes that the current inter-purchase time (ipt) between $t_{x,i}$ and $t_{x+1,i}$ corresponds to the mean of the customer's previously observed ipts. As there are no observed ipts for single buyers, they have no individual heuristic and are assigned the mean ipt of all repeat buyers instead. This leads to the following framework:

$$t^{heur}_{x+1,i} = \begin{cases} t_{x,i} + M_i(ipt), & t_{x,i} + M_i(ipt) > T_i \\ T_i + M_i(ipt), & t_{x,i} + M_i(ipt) \le T_i \end{cases} \tag{9}$$

with

$$M_i(\mathrm{ipt}) = \begin{cases} \frac{t_{x,i}}{x_i}, & x_i > 0 \\ \frac{1}{\sum_{j=1}^{n} 1_{x_j>0}} \cdot \sum_{j=1}^{n} \frac{t_{x,j}}{x_j} \cdot 1_{x_j>0}, & x_i = 0. \end{cases} \tag{10}$$

The first case reflects the general idea of the heuristic. The second case ensures that the predicted purchase does not take place before the beginning of the forecast period. As the exponential distribution is right skewed, it makes sense to use quantile- rather than mean-based measures to reduce the impact of outliers.

3 Data set framework

3.1 Simulated data

The behavioural characteristics of the 3000 simulated data sets are based on the estimates of existing empirical data sets and are shown in Table 2. The scope of the cohort size n and the length of the calibration period T (in weeks) meet the small sample requirements from Hoppe and Wagner (2010) and Simon and Adler (2022).

Table 2 Behavioural and Operational Characteristics for the Simulation Study

Behavioural characteristics	Operational characteristics
$E(\lambda) \in [0.02; 0.3]$ $CV(\lambda) \in [0.5; 2.5]$ $E(\mu) \in [0.02; 0.2]$ $CV(\mu) \in [0.5; 2.5]$	$N \in [1{,}000; 4{,}000]$ $T \in [26; 72]$

Note: CV denotes the coefficient of variation

All these values were drawn from uniform distributions within the intervals as specified in Table 2. Each customer's initial purchase occurred randomly within the first 90 days.

The behavioural characteristics consider a wide range of purchase and dropout processes. The minimum value of $E(\lambda)=0.02$ implies that an average customer (i.e., a customer whose purchase parameter λ equals the mean of the distribution which is 0.02) has a mean ipt of $\frac{1}{0.02}=50$ weeks which implies a very low purchase frequency in the data set. A data set with a high purchase frequency (i.e., where $E(\lambda)=0.3$) implies that an average customer makes a purchase every $\frac{1}{0.3}=3.33$ weeks. In the dropout process, the mean lifetime of an average customer ranges from $\frac{1}{0.2}=5$ weeks to $\frac{1}{0.02}=50$ weeks.

The coefficients of variation $CV(\lambda)$ and $CV(\mu)$ describe the heterogeneity of the parameters λ and μ across customers. For their smallest value of 0.5, this means that the values of the corresponding individual parameters deviate on average by 0.5 = 50% from their mean value. The maximum value of 2.5 indicates an average deviation from the mean value of 2.5 = 250%.

The cohort size N and the length of the calibration period T are randomly chosen from the intervals shown in Table 2. Independent from the length of the calibration period, each data set is assessed for three different lengths of the forecast period $T^* \in \{13, 26, 52\}$ weeks.

3.2 Empirical data sets

The following four empirical data sets are used to validate the results of the simulation study whose operational characteristics are summarised in Table 3:

- *CDNow* contains purchase data of 2,357 customers of an online CD retailer from January 1997 to June 1998 that were acquired in the first quarter of 1997. This data set is the most common benchmark for CBA models and has been studied extensively (Jerath et al. 2011; Bemmaor and Glady 2012; Fader and Hardie 2001; Platzer and Reutterer 2016). The data set covers a time span of 78 weeks, half of which are used as the calibration period. As a maximum of 39 weeks remain for the forecast period, only the first two of the forecast periods T* of 13, 26 and 52 weeks used in the simulation study are available here for validation.

Table 3 Operational characteristics of the empirical data sets

	Cohort size	calibration period (weeks)	available forecast periods (weeks)
CDNow	N = 2,357	T = 39	$T^* \in \{13, 26\}$
Grocery	N = 1,525	T = 52	$T^* \in \{13, 26, 52\}$
Gusto	N = 1,000	T = 78	$T^* \in \{13, 26, 52\}$
Homeshopping	N = 1,000	T = 78	$T^* \in \{13, 26, 52\}$

- *Grocery* is a data set from an online grocery retailer including purchase data from 1,525 customers who made their initial purchase in the first quarter of 2006 and has a total observation period of 2 years, 52 weeks of which are used as a calibration period. Hence, all three standard forecast periods T* ∈ {13, 26, 52} can be applied here. This data sets was also used by Platzer and Reutterer (2016).
- *Gusto* is a proprietary data set which includes dates of 3,510 customers of an online shop who were acquired between the beginning and the end of 2006. This study only use an excerpt of 1000 customers here. As the total time span covered by this data set is five years, all standard forecast periods T* ∈ {13, 26, 52} can be used here after a calibration period of 78 weeks. In contrast to the Grocery data set, this data sets is related to tobacco product purchases which occur less frequently.
- *Homeshopping* is an excerpt of a proprietary data set and contains orders from a home shopping TV channel of 1000 customers from July 2004 until May 2009. The total time span of almost 5 years enables the analysis of all standard forecast periods T* ∈ {13, 26, 52} after a calibration of 78 weeks. The nature of this data set is different to the others as it typically reflects a broader range of product categories, often involves higher engagement through televised promotions, and may exhibit different purchasing patterns, such as impulse buying and repeat orders driven by limited time offers.

3.3 Parameter estimation

The MLE estimates are determined using the CLVTools package (Bachmann et al. 2022). To determine confidence intervals for MLE, 500 bootstrapping samples were drawn for each data set.

The MCMC procedure is based on Abe (2009) and is provided in the BTYDplus package (Platzer 2021a). Every estimation uses the hyper prior parameters as derived by Simon and Adler (2022) and comprises 10,000 MCMC draws after a burn-in period of 2000 draws. Using four chains and after a thinning of 20 (i.e., storing every 20th draw), the final estimates contain 2000 values for each individual and heterogeneity parameter of each data set.

4 Results

4.1 Purchase forecast

The four methods of determining the individual purchase forecast (SPP, ICE, MLE, heuristic) as described in 2.1 are applied to the three prediction period lengths T* ∈ {13, 26, 52}. As already stated, the median of the MCMC forecasts is used as point estimate. In addition to the MLE point estimate, the MLE bootstrapping samples are used to provide a distribution of the purchase forecast from which the

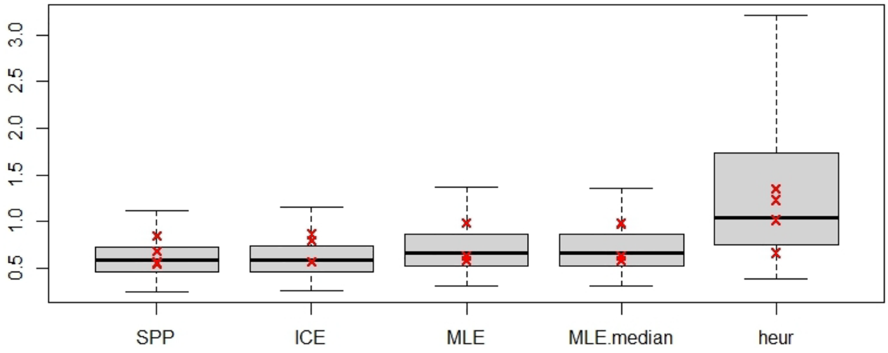

Fig. 1 Normalised mean absolute error (nMAE) of the individual purchase forecast (T* = 26 weeks). *Note:* The crosses represent the results of the four empirical data sets

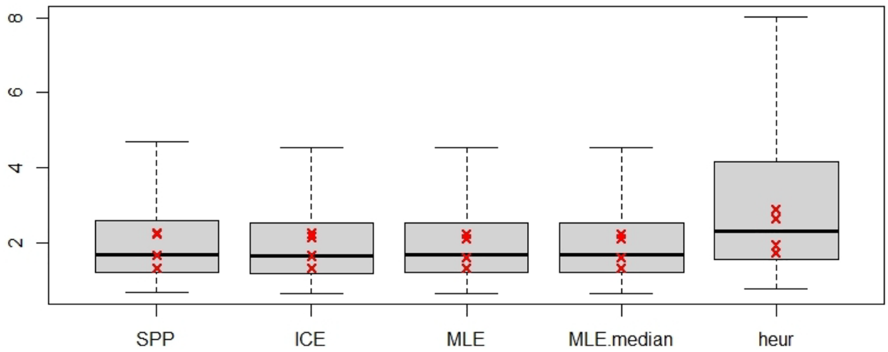

Fig. 2 Normalised root mean squared error (nRMSE) of the individual purchase forecast (T* = 26 weeks)

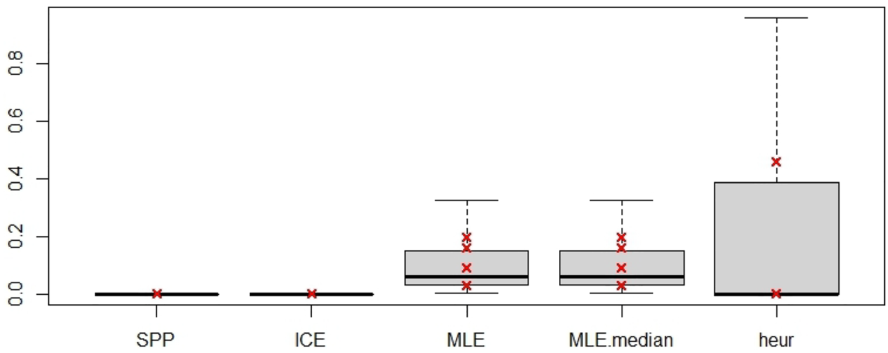

Fig. 3 Normalised median absolute percentage error (nMdAE) of the individual purchase forecast (T* = 26 weeks)

median is taken as a second MLE based estimate (named MLE.median). The idea is to examine whether this bootstrapped median leads to significantly different (or better) results than the prediction from the classical MLE prediction.

Figures 1, 2 and 3 show the deviation measures obtained for the different forecasting methods. As the plots are very similar for the different lengths of the forecast period T*, only the results for T* = 26 weeks are presented here for reasons of clarity. The boxplots represent the results from the simulated data, the red crosses represent the results from the four real-world data sets. A first finding is that all deviation values of all four empirical data sets lie within the 95% confidence intervals resulting from the simulated data sets, i.e., within the boxplots.

The MCMC forecasts SPP and ICE in Fig. 1 show a slightly better performance in the nMAE than MLE for the simulated data sets which is not considerable enough to be confirmed by the empirical data sets. The nRMSE (Fig. 2) hardly shows any difference between the parameter-based predictions but a larger scale of deviation for the heuristic which also holds for the empirical data sets. The nMdAE in Fig. 3 displays a very unusual pattern as the values are close or equal to zero across all data sets for SPP and ICE. The reason for this is that the MCMC methods use the augmented lifetimes $\{\tau_i\}_{i=1,\dots,n}$ which allows them to obtain a purchase forecast of exactly zero. Since it is very common for CBA data sets that a majority of the customers make no purchase in the forecast period, the correct assignment of these inactive customers causes a large set of zero deviations with a median being close to zero. Even though this outperformance emphasises the advantage of the MCMC predictions regarding zero buyers, the results should not be overrated as they are mainly caused by a group which is not in the focus of marketing activities.

Neither of the plots show a visible difference between MLE and MLE.median. Figure 4 displays the standard deviations of the individual purchase predictions for the MCMC methods and the bootstrapped MLE for the real-world data sets. Unlike MCMC, a bootstrapped sample yields the confidence interval of the MLE rather than of the posterior distribution which makes the confidence intervals much smaller. In fact, this leads to the bootstrapped median and mean forecasts being so close to the forecast of the MLE point estimate that these are essentially indistinguishable from each other.

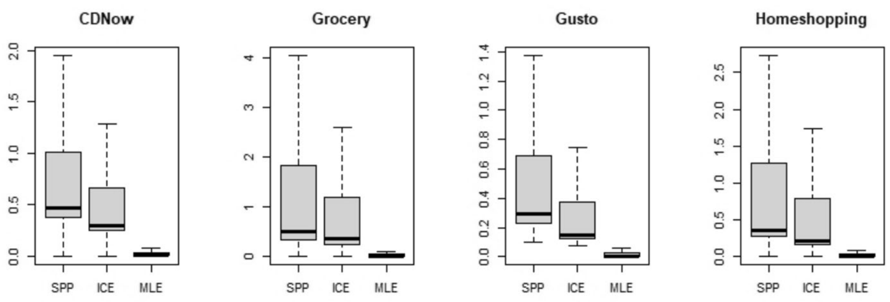

Fig. 4 Standard deviations of the individual purchase forecasts (T* = 26 weeks)

In the following analyses, therefore, only the forecast based on the MLE point estimate is taken into account, as this would otherwise increase the complexity of the presentations, but it is not expected to yield any new conclusions.

4.2 Active customer identification

Figure 5 shows the sensitivity and precision as well as the overall accuracy of the correctly assigned customers. As MCMC tends to underestimate x*, SPP and ICE have a rather low sensitivity. This implies that they tend to classify active customers as inactive. On the other hand, customers that are predicted active by MCMC are very likely to be assigned correctly, leading to high precision values. Taking these effects together, the MCMC methods yield the highest ratio of correctly assigned customers. The poor sensitivity of the CDNow dataset (29.4% for SPP and 42.8% for ICE) can be explained by the fact that 12.6% of the customers in this data set make exactly one purchase in the forecast period. This is a very high ratio that is only exceeded by 3% of the simulated data sets. Hence, the underestimation of the MCMC methods causes a misclassification of many active customers as being inactive.

Until now, customers were classified as active if their future purchase forecast was above the default value of c = 0.5. Although this threshold is a very intuitive cut-off value, it does not automatically make it the best choice for every data set. As there is a trade-off between sensitivity and precision, meaning that improving one often comes at the cost of the other, the only value that can reasonably be maximised by varying the value of c is the overall accuracy. Figure 6 displays

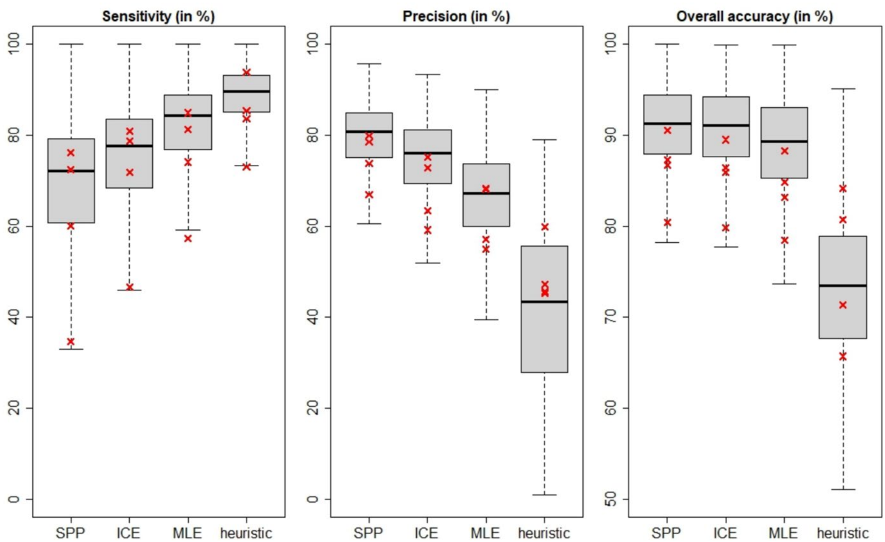

Fig. 5 Performance Measures for the active customer identification (T* = 26 weeks). *Note:* The crosses represent the performance measures of the four empirical data sets

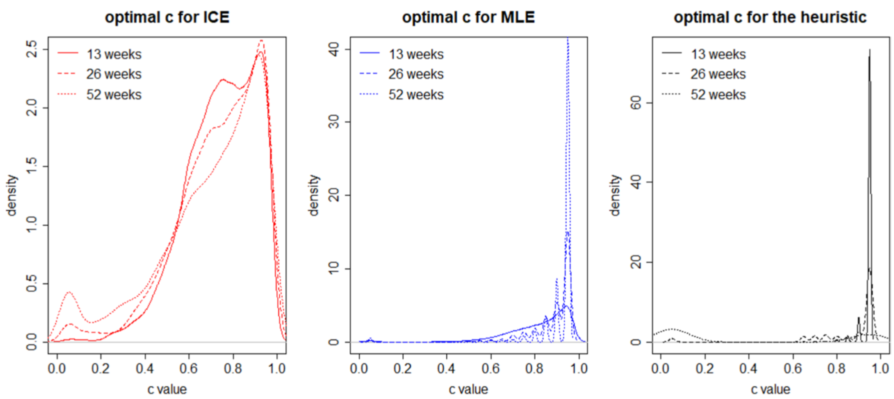

Fig. 6 Optimal cut-off values for the simulated data sets

how this optimised cut-off value c is distributed across the simulated data set. The optimal c value not only depends on the data set itself, but also on the length of the forecast period, which makes the best choice of the cut-off unmanageable for new data sets.

SPP can play out another strength here: as it yields an integer purchase forecast and hence a (mostly) integer median, this is the only method for which varying the cut-off between 0 and 1 is unnecessary. The actual ratio of correctly assigned customers for c = 0.5 and the optimal c value (in italic) in Table 4 indicate that SPP mostly outperforms even the optimised allocation of the other methods. The variability in the optimal c values also holds for the empirical data sets: for the same data set, the ICE cut-off varies by up to 0.2, for MLE by up to 0.15 and for the heuristic even by up to 0.9. It is therefore recommendable to apply SPP for predicting customer activity.

Table 4 Correctly assigned Customers for T* = 26 weeks and c = 0.5

	SPP	ICE	MLE	heuristic
CDNow	80.4% (80.5%)	79.8% (80.4%)	78.4% (80.1%)	71.4% (79.0%)
Grocery	86.7% (86.7%)	85.9% (87.0%)	83.1% (86.1%)	65.7% (76.1%)
Gusto	90.5% (90.6%)	89.5% (91.5%)	88.3% (91.1%)	84.1% (87.1%)
Homeshopping	87.3% (87.3%)	86.4% (87.1%)	84.8% (86.6%)	80.7% (82.9%)

Note: The values written in italic show the ratio of the correctly assigned customers when applying the optimal cut-off value c

4.3 Future Top A% customers

Since customer prioritisation is a concept for longer-term customer relationships, using the forecasts of the 52-week horizon for the analysis is most reasonable. As pointed out in 2.3, this study only considers those data sets that have at least 10% active customers for the Top 10% and at least 20% active customers for the Top 20% segment to ensure that the corresponding quantile of the future purchases is greater than zero. Hence, 95% of the considered simulated data sets have no more than 12.3% customers belonging to the Top 10% share due to multiple values (35.4% for the Top 20%). Figure 7 reveals that there is no significant difference between the three model-based classifications, but they all outperform the heuristic.

For three quarters of the simulated data sets, between 63 and 78% of the Top 10% and between 69 and 81% of the Top 20% customers are classified correctly. The lower boundary of the Top 20% identification is above the value determined in Huang's (2012) simulation study where he compared MLE and heuristics, each with and without covariates. This can be due to multiple reasons like different behavioural or operational characteristics of the data sets or the fact that he uses the empirical data set generating parameters instead of parameter estimates.

Taking a closer look at the empirical data reveals that the CDNow and the Gusto data sets are barely suitable for this type of analysis. CDNow has a low mean purchase frequency of 0.03 and a maximum T* of 39 weeks which gives customers hardly any chance to differentiate from each other. The mean purchase rate of the Gusto data set is even smaller with 0.02. Hence, the CDNow Top 20% include all active customers (i.e., with $x^* > 0$) and the Gusto Top 20% are even equal to the whole customer base since only 17.4% of the customers make a purchase in (T, T + T*]. Prolonging the prediction period for the Gusto data set from 52 to 104 weeks still lets every active customer join the Top 20% segment. However, assigning every (active) customer to the Top 20% of the customers certainly does not meet the idea behind this type of customer segmentation.

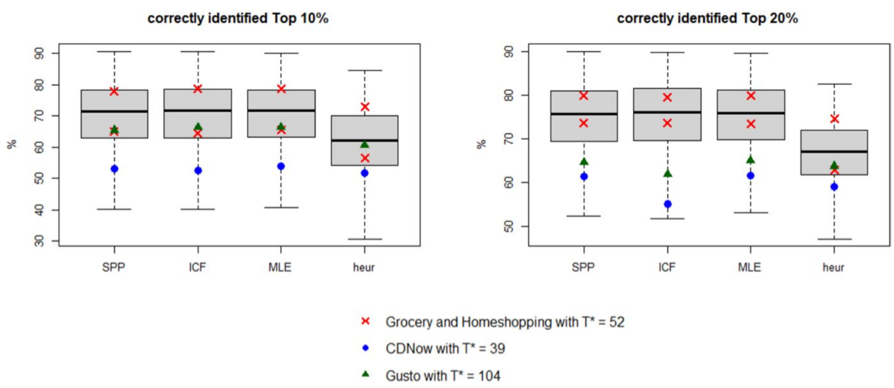

Fig. 7 Correctly identified top customers for T* = 52 weeks

Despite their lack of suitability, the results for CDNow (with $T^* = 39$ weeks) and Gusto (with $T^* = 104$ weeks) are also displayed in Fig. 7 as circles and triangles, respectively.

4.4 Timing of the next purchase

The shape of the exponential distribution already suggests that predicting drawn ipts is a difficult undertaking. The right skew of the distribution tends to cause high overestimations for t_{x+1}. In contrast to this, the heuristic has an upper boundary of $(T_i + T)$.

Figure 8 displays the ratio of active customers whose forecast for t_{x+1} deviates less than a given threshold (2, 4, and 8 weeks) from its true value. The plots show that the heuristic slightly outperforms MCMC and MLE with increasing deviation interval. For the empirical data sets, the predictions work poorly, in particular for the CDNow and the Gusto data set. The high number of single buyers in these data sets impede a reliable forecast of their next purchase timing as there are no observable ipts for them. In the CDNow data set, this even holds for 22.9% of the active customers.

It is certainly worth discussing whether it makes sense to rely on a purchase timing forecast that, in most cases, achieves a maximum accuracy of four weeks for only 60% of the active customer base. The optimistic results from Korkmaz et al. (2013) cannot be confirmed here. The reason for this is that their study design also includes the inactive customers who make no future purchase. Their contribution to the accumulated deviation measures is very small, leading to a biased deviation figure which is not representative for the active customers, even though these are the only ones of interest.

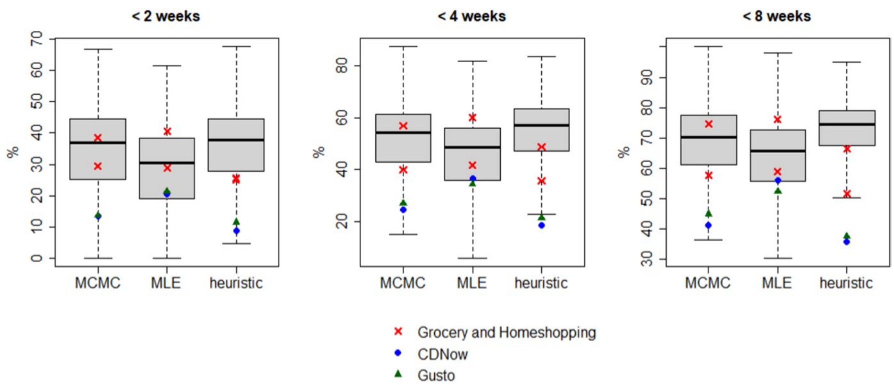

Fig. 8 t_{x+1} Forecast deviations

5 Discussion and implications

This study is the first generalised effort to compare different Pareto/NBD model-based forecasts with heuristics, aiming to give practitioners guidelines for the predictive power of the different forecasting methods. Previous contributions mainly relied on a handful of empirical data sets (Korkmaz et al. 2013; Batislam et al. 2007; Wübben and Wangenheim 2008), claiming that heuristics yield competitive results to model-based approaches. However, the analysis of the presented simulation study shows that the heuristics are less accurate than the model-based forecasts when predicting the future number of purchases or identifying either the future active customers or the top segment of the customer base. Still, the analysis of the next purchase timing prediction reveals that not all types of predictions that can technically be performed make sense. For all types of forecasts, different MCMC- and MLE-based methods were compared to identify the best practice.

Firstly, regarding the individual purchase forecast x_{fc}*, the deviation measures (Figs. 1, 2, 3) reveal that the parametric predictions outperform the heuristic in both the simulated and the empirical data sets. The nRMSE are very similar across the MCMC and MLE based predictions, whereas the nMdAE shows values close to or equal to zero for the MCMC methods because of their ability to assign purchase forecasts of zero to customers. The nMAE shows a slightly better performance of MCMC for the simulated data which is not confirmed by the empirical data sets.

Secondly, the identification of active customers shows that MCMC estimates achieve higher ratios of correctly classified customers than MLE and the heuristic. In opposite to the previous approach in literature, which is based on the non-observable P(alive) (Batislam et al. 2007; Wübben and Wangenheim 2008), this study refers to actual purchases which can be validated in practice. This enables managers to enhance their understanding of individual customer behaviour and tailor their marketing efforts accordingly. It is recommendable to apply the SPP allocation because it dispenses with the cumbersome selection of the cut-off value c.

Thirdly, the study of future Top A% customers reveals that all model-based assignments (SPP, ICE, and MLE) perform very similarly, outperforming the heuristic approach. The analysis also shows that practitioners should take a close look at whether their data set is appropriate for this type of customer segmentation. Short calibration periods and/or very low purchase rates within a data set may produce too little variation in the purchase data to allow a sensible differentiation of the customer base. Prolonging the calibration period or using additional information like the individual customers' revenue might help to receive substantial differences between the customers, although the latter recommendation was not in the scope of this study. The CDNow and Gusto data sets are good examples for such unsuitable data sets. Since the idea behind a special treatment of the Top A% customers is to achieve a substantial increase in the overall revenue by applying little effort only, incentivising literally all active customers as they belong to the top segment of the customer base is contradictory to the idea of disproportionate marketing.

Finally, the prediction of the timing of the next purchase t_{x+1} remains a challenging task as the quality of the forecast itself is rather daunting. In cases where marketing managers base their incentives and mailings on a variety of information, forecasting can provide an additional value, but it should not be used as a stand-alone marketing tool.

In summary, this study proves the superiority of Pareto/NBD based forecasts over heuristics which is contradictory to previous results (Wübben and Wangenheim 2008; Huang 2012). It also shows that MCMC provides better predictions of future purchases and identification of active customers than MLE. The differences may not be very large in themselves, but MCMC provides not only a point estimate for each forecast, but the entire posterior distribution, so that uncertainties in the forecasts can also be represented without additional effort. Even though bootstrapping can be applied to also generate forecast intervals from MLE, it only accounts for uncertainty in the true value of the likelihood's mode but not in the true parameter value which limits the additional benefit.

Despite the valuable insights and significant findings obtained from this study, it is essential to acknowledge certain limitations that may impact the interpretation and generalisation of the results. This study tried to cover a reasonable setting of parameter spaces for the simulated data sets. Still, customer behaviour can vary significantly across different sectors, product types, and demographics. Thus, the results might not be directly applicable to all scenarios. The error measures and their quantiles presented here should therefore be interpreted as indicators rather than hard limits. Nevertheless, all basic statements of the simulated data sets are supported by the empirical data sets.

It is certainly worth discussing whether the standard Pareto NBD model, which does not include covariates or a spending process, is the right data basis for marketing decisions. Especially in order to measure the effect of marketing measures, for the optimisation of which these models are used, the consideration of covariates is indeed essential. Still, understanding the strengths and weaknesses of the fundamental model is crucial before adding additional complexity which is left for further research. Nevertheless, the results show that even with the basic model examined here, classification and prioritisation decisions are significantly better than with the heuristic.

Since forecasting is the actual goal of any modelling activity in CBA, investigating the quality of these predictions for different models (e.g., BG/NBD or Pareto/GGG) or model variants (like incorporating covariates) is another natural extension of this study. A methodically different but more complex variant of this study could consider the CLV instead of the number of purchases.

Appendix

MCMC algorithm for the Pareto/NBD model

The MCMC algorithm used here is based on Abe (2009). He introduced it to estimate an extended Pareto/NBD model which relaxes the interdependency assumption of the purchase and dropout process. Nevertheless, Platzer (2021a, b) adapted this procedure to the original Pareto/NBD model in the BTYDplus package which is the version I used in this paper. The algorithm applies data augmentation techniques (Tanner & Wong 1987) to determine estimates for the individual lifetime $(\tau_i)_{i=1,\ldots,n}$. This additional information simplifies the posteriors to Gamma distributions that I can directly draw from.

Let x_i denote the number of repeat purchases that customer i makes within the observation period [0, T]. As customers start their relationship with the firm at different dates, the individual observation period is T_i with $t_{x,i}$ being the timing of the last purchase before T_i.

In this algorithm, the following steps are carried out:

1. Initialise the parameters $\{\lambda_i, \mu_i, \tau_i\}_{i=1,\ldots,n}$ and $\{r, \alpha, s, \beta\}$.
2. Draw the individual purchase parameters $\{\lambda_i\}_{i=1,\ldots,n}$ from $\lambda_i \sim \Gamma(x_i + r, \alpha + \min(\tau_i, T_i))$.
3. Draw the individual lifetime parameters $\{\mu_i\}_{i=1,\ldots,n}$ from $\mu_i \sim \Gamma(s + 1, \beta + \tau_i)$.
4. Draw the indicator $\{z_i\}_{i=1,\ldots,n}$ which denotes whether a customer is still alive in $(\tau_i,)_{i=1,\ldots,n}$ using.
5. $P(z_i = 1) = P(\tau_i > t_i \mid \lambda_i, \mu_i, t_{x,i}, T_i)$ and.
6. $P(z_i = 0) = 1 - P(z_i = 1)$.
7. Draw $\{\tau_i\}_{i=1,\ldots,n}$ from an exponential distribution if $z_i = 1$ and from a double truncated exponential distribution if $z_i = 0$:
8. $\tau_i \sim T_i + \mathrm{rexp}(\mu_i)$ if $z_i = 1$ and $\tau_i \sim -\frac{\ln\left[(1-\mathrm{runif})\cdot e^{-(\lambda_i+\mu_i)\cdot t_{x,i}} + \mathrm{runif}\cdot e^{-(\lambda_i+\mu_i)\cdot T_i}\right]}{\lambda_i+\mu_i}$ if $z_i = 0$.
9. Using slice sampling, draw$\{r, \alpha\}$ simultaneously from

$$\prod_{i=1}^{n} c.$$

10. Using slice sampling, draw $\{s, \beta\}$ simultaneously from

$$s^{h_5-1}e^{-sh_6} \cdot \beta^{h_7-1}e^{-\beta h_8} \prod_{i=1}^{n} \frac{\beta^s \mu_i^{s-1} e^{-\mu_i \beta}}{\Gamma(s)}.$$

11. Repeat from step 2 with updated parameter values.

Estimation procedures for the future purchase

For simplification reasons, the formulas are presented for a single customer and refrain from noting the customer index i. I use the following procedures for determining the purchase forecast:

Simulated purchase patterns (SPP)

Simulating the customer's future purchase pattern requires estimates for {λ, τ}. For each MCMC draw j (j = 1,…,J) where the customer is still alive in T (i.e., $\tau_j > T$), I draw inter-purchase times $\{\vartheta_{j,1},\ldots,\vartheta_{j,k+1}\}$ from Exp(λ) until the dropout time or the end of the prediction period is exceeded, i.e. $T + \vartheta_{j,1} + \cdots + \vartheta_{j,k} \le \min(T + t^*, \tau)$ and $T + \vartheta_{j,1} + \cdots + \vartheta_{j,k+1} > \min(T + t^*, \tau)$.

The purchase forecast for the j-th draw is then given by

$$x^*_{fc,j} = \begin{cases} \vartheta_{j,k} & \text{if} \tau_j > T \\ 0 & \text{if} \tau_j \le T. \end{cases}$$

Individual conditional expectation (ICE)

Fader & Hardie (2005) provide a formula for the expected number of future purchases given that the customer is still alive in T:

$$E\left[x^*|\lambda, \mu, T, t^*, \tau > T\right] = 1_{\tau>T} \cdot \frac{\lambda}{\mu}\left[1 - e^{-\mu t^*}\right]. \tag{11}$$

Applying this formula requires the individual MCMC estimates for $\{\lambda_j, \mu_j, \tau_j\}_{j=1,\ldots,J}$.

Heterogeneous closed form

In case of MLE where estimates for {r, α, s, β} are provided only, I apply the heterogeneous conditional expectation (Schmittlein et al. 1987) given as

$$\begin{aligned} E\left[x_i^*|r,\alpha,s,\beta,x,t_x,T,t^*\right] = {} & \left[\frac{\Gamma(r+x)\alpha^r\beta^s}{\Gamma(r)(\alpha+T)^{r+x}(\beta+T)^s} / L\left(r,\alpha,s,\beta|x,t_x,T\right)\right] \\ & \times \frac{(r+x)(\beta+T)}{(\alpha+T)(s-1)}\left[1 - \left(\frac{\beta+T}{\beta+T+t^*}\right)^{s-1}\right]. \end{aligned} \tag{12}$$

Additional results like the cumbersome formula for $L(r, \alpha, s, \beta \mid x, t_x, T)$ can be found in Fader & Hardie (2005).

Heuristic

The heuristic assumes that every customer keeps buying at his or her previous purchase rate. Since single buyers do not provide a provide a purchase rate, their prediction is set to zero, yielding:

$$x^*_{heur,i} = \begin{cases} \frac{x_i}{T_i} \cdot T^* & , x_i > 0 \\ 0 & , x_i = 0 \end{cases}. \tag{13}$$

Estimation of the timing of the next purchase using MLE

For clarity reasons, we omit the customer index i in the following formulas. The timing of the next ipt (t_{x+1} – T) follows an exponential distribution

$$f\left(t_{x+1} - T|\tau > T, \lambda\right) = \lambda e^{-\lambda(t_{x+1}-T)}. \tag{14}$$

The posterior distribution of λ is given by

$$g(\lambda|x, \tau > T, r, \alpha) = \frac{\lambda^x e^{-\lambda T} \frac{\lambda^{r-1}\alpha^r e^{-\lambda\alpha}}{\Gamma(r)}}{\int_0^\infty \lambda^x e^{-\lambda T} \frac{\lambda^{r-1}\alpha^r e^{-\lambda\alpha}}{\Gamma(r)}} = \frac{(\alpha + T)^{x+r} \lambda^{x+r-1} e^{-\lambda(\alpha+T)}}{\Gamma(x+r)}. \tag{15}$$

Bringing (*14*) and (*15*) together to determine the heterogeneous density of the next ipt yields

$$\begin{aligned} f\left(t_{x+1} - T|x, \tau > T, r, \alpha\right) &= \int_0^\infty f\left(t_{x+1} - T|\tau > T, \lambda\right) g(\lambda|x, \tau > T, r, \alpha) d\lambda \\ &= \frac{(\alpha + T)^{x+r}}{\Gamma(x+r)} \int_0^\infty \lambda^{x+r} e^{-\lambda(\alpha+t_{x+1})} d\lambda = \frac{(\alpha + T)^{x+r}(x+r)}{\left(\alpha + t_{x+1}\right)^{x+r+1}}. \end{aligned} \tag{16}$$

The expected value is then given by

$$\begin{aligned} E\left(t_{x+1} - T|x, \tau > T, r, \alpha\right) &= \int_0^\infty \left(t_{x+1} - T\right) \frac{(\alpha + T)^{x+r}(x+r)}{\left(\alpha + t_{x+1}\right)^{x+r+1}} d(t_{x+1} - T) \\ &= -(\alpha + T)^{x+r} \left[\left(\alpha + t_{x+1} + T\right)^{-(x+r)} \cdot \left((t_{x+1} - T) \frac{\alpha + t_{x+1}}{x + r - 1} \right) \right]_0^\infty \end{aligned} \tag{17}$$

and converges to

$$\frac{\alpha + T}{x + r - 1} \tag{18}$$

if and only if $x + r > 1$. This implies that the expected value of the timing of the next purchase cannot be determined for single buyers (i.e., customers with $x = 0$) in a data set with estimated $r < 1$.

The median Q_{50} can be determined using (16)

$$\begin{aligned} \int_0^{Q_{50}-T} \frac{(\alpha + \mathrm{T})^{\mathrm{x+r}}(\mathrm{x+r})}{\left(\alpha + \mathrm{t_{x+1}}\right)^{\mathrm{x+r+1}}} \mathrm{d}\left(\mathrm{t_{x+1}} - \mathrm{T}\right) &= (\mathrm{x+r})(\alpha + \mathrm{T})^{\mathrm{x+r}} \int_{\mathrm{T}}^{Q_{50}} (\alpha + \mathrm{t}(\mathrm{x}+1)^{-(\mathrm{x+r})-1} \mathrm{dt_{x+1}} \\ &= 1 - \frac{(\alpha + \mathrm{T})^{\mathrm{x+r}}}{\left(\alpha + Q_{50}\right)^{\mathrm{x+r}}} = 0.5 \iff Q_{50} = 2^{\frac{1}{\mathrm{x+r}}}(\alpha + \mathrm{T}) - \alpha. \end{aligned} \tag{19}$$

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Data availability The codes used to receive the presented results are publicly available at https://github.com/PNBDForecast/PNBD-Forecast. The simulated data sets and estimates are not provided due to their size but can be reproduced by using the provided code. Because of data use restrictions, the real data sets are only available on request.

Declarations

Conflict of interests The author has no relevant financial or non-financial interests to disclose.

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